

CPEN 416 / EECE 5710

Lecture 20: Correcting arbitrary single-qubit errors; intro to stabilizers

Thursday 20 November 2025

Announcements

- Quiz 9 on Tuesday
- Last tutorial assignment on Monday
- Reminder to fill out weekly surveys for project

Last time

We discussed the motivation for quantum error correction, and developed a mathematical (/graphical) model for noise.

Last time

We saw our first quantum error correcting code: the quantum repetition code, or bit flip code.

Last time

We designed logical operations for the bit flip code.

Last time

We designed a circuit to detect if a bit flip error occurred, and identify its location.

Learning outcomes

Today:

- ➊ Correct bit *and* phase flip errors with the 9-qubit code
- ➋ Outline the conditions under which errors can be corrected
- ➌ Define the stabilizers of a quantum error correcting code

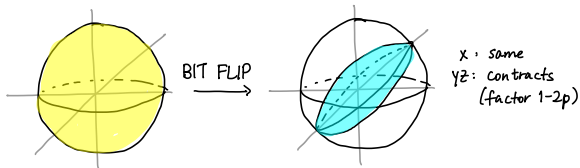
Bit flip code: recovery operation

Design a circuit to *recover* the state of the first qubit

Bit flip code: full circuit for bit flip code

Bit flip code: analysis

Imagine any possible state that can go through the channel:



Exercise: Determine $F(|\psi\rangle, \Phi(|\psi\rangle \langle\psi|))$ where Φ is the *bit flip channel* with parameter p .

Bit flip code: analysis

Compare overall fidelity of the state *after the error*, vs. fidelity *after recovery*.

After the error

After recovery, have the state

Phase flip errors

With our encoding

and appropriate circuitry, we can correct *bit flip errors* but not phase flip errors.

Phase flip code: encoding circuit

Main idea: make phase flip errors “look like” bit flip errors.

Shor code

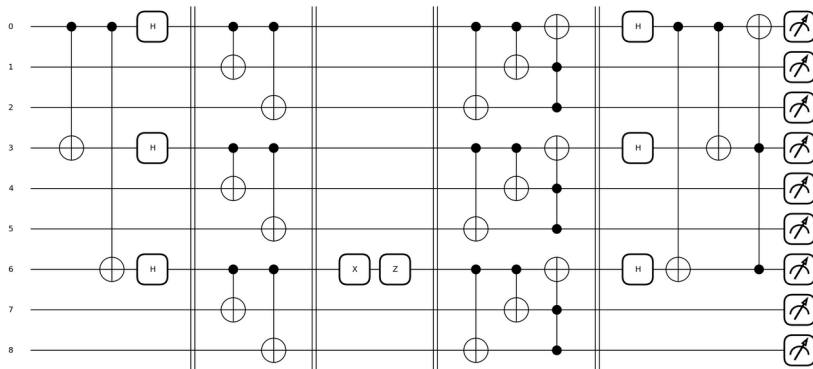
To correct a combination of one bit flip and/or phase flip error, we can *concatenate* codes.

Shor code

The Shor code can correct one *arbitrary error on a single qubit*.

Imagine a bit + phase flip on qubit 6:

Shor code



Correcting arbitrary errors

The 9-qubit code can correct *arbitrary* errors on a *single qubit*.

Represent possible errors as $\{E_i\}$, full error channel as

We can correct $\{E_i\} = \{I, X_j, Z_j, X_j Z_j\}_j$. If we can write an arbitrary error $\mathcal{E}_k$ on qubit j as

then we can correct it too.

Conditions for quantum error correction

How do we know if a particular error is correctable by a QECC?

Formal definition of a quantum error correcting code is a *subspace*, C , called the **codespace**.

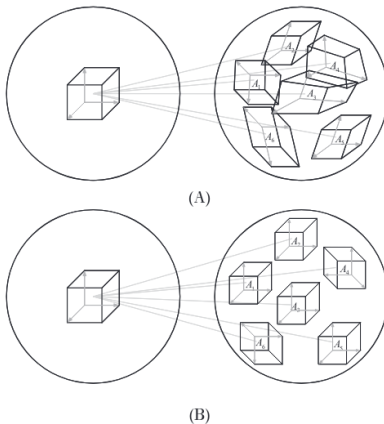
Example: bit flip code.

- Codewords:
- Codespace: contents have the form

Define a **projector** onto the codespace,

Conditions for quantum error correction

The errors this code corrects are all mapped to different **orthogonal** subspaces of the full Hilbert space.



Conditions for quantum error correction

Theorem 10.1: (Quantum error-correction conditions) Let C be a quantum code, and let P be the projector onto C . Suppose $\mathcal{E}$ is a quantum operation with operation elements $\{E_i\}$. A necessary and sufficient condition for the existence of an error-correction operation $\mathcal{R}$ correcting $\mathcal{E}$ on C is that

(10.16)

for some Hermitian matrix α of complex numbers.

If such an $\mathcal{R}$ exists, $\{E_i\}$ is called a *correctable set of errors*.

Correctable errors: bit flip code

Consider $\{I, E_1 = XII, E_2 = IXI, E_3 = IIX\}$

Try including a 2-qubit error:

Example: amplitude damping

Not immediately clear how to express this in terms of no errors, bit, phase or bit/phase flip errors. Can we still correct it?

Straightforward to re-express K_1 :

For K_0 ,

Bit flip code: recovery revisited

We've designed two recovery circuits for the bit flip code.

Stabilizers

We can consider a more general invariant than parity: eigenvalues w.r.t. special subsets of the Pauli group.

The n -qubit Pauli group is

Example: Which two-qubit Paulis is this Bell state a $+1$ eigenstate of?

Bit flip code: stabilizers

Consider our logical states:

Which three-qubit Paulis are these states $+1$ eigenstates of?

Stabilizers

Let S be a subgroup of $\mathcal{P}_n$.

Let V_S be a set of states that are $+1$ eigenstates for all $P \in S$.

Then, S is the **stabilizer** of V_S .

Facts about S :

- $-I$ is never in S
- all items of S commute
- choosing S uniquely defines the fixed subspace, V_S

Bit flip code: stabilizers generators

Let's determine a *minimal* representation of the group in terms of its **stabilizer generators**.

Bit flip code: stabilizer measurement

We can use this formalism to construct new circuits for error detection and recovery: simply *measure the stabilizer generators*.

Next time

Next class:

- More stabilizer codes: phase flip, $[[5, 1, 3]]$ code, Steane code

Action items:

- 1 Quiz 9
- 2 Work on project

Recommended reading:

- From this class: Codebook EC; 10.3, 10.5
- For next class: Codebook EC; 10.1-10.5